

Graph & Geometric Deep Learning for Drug Toxicity Prediction

A scaffold-split benchmark of fingerprints, ChemBERTa, 2D GNNs, 3D SchNet, and multitask learning across five toxicity tasks.

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CORE FINDING

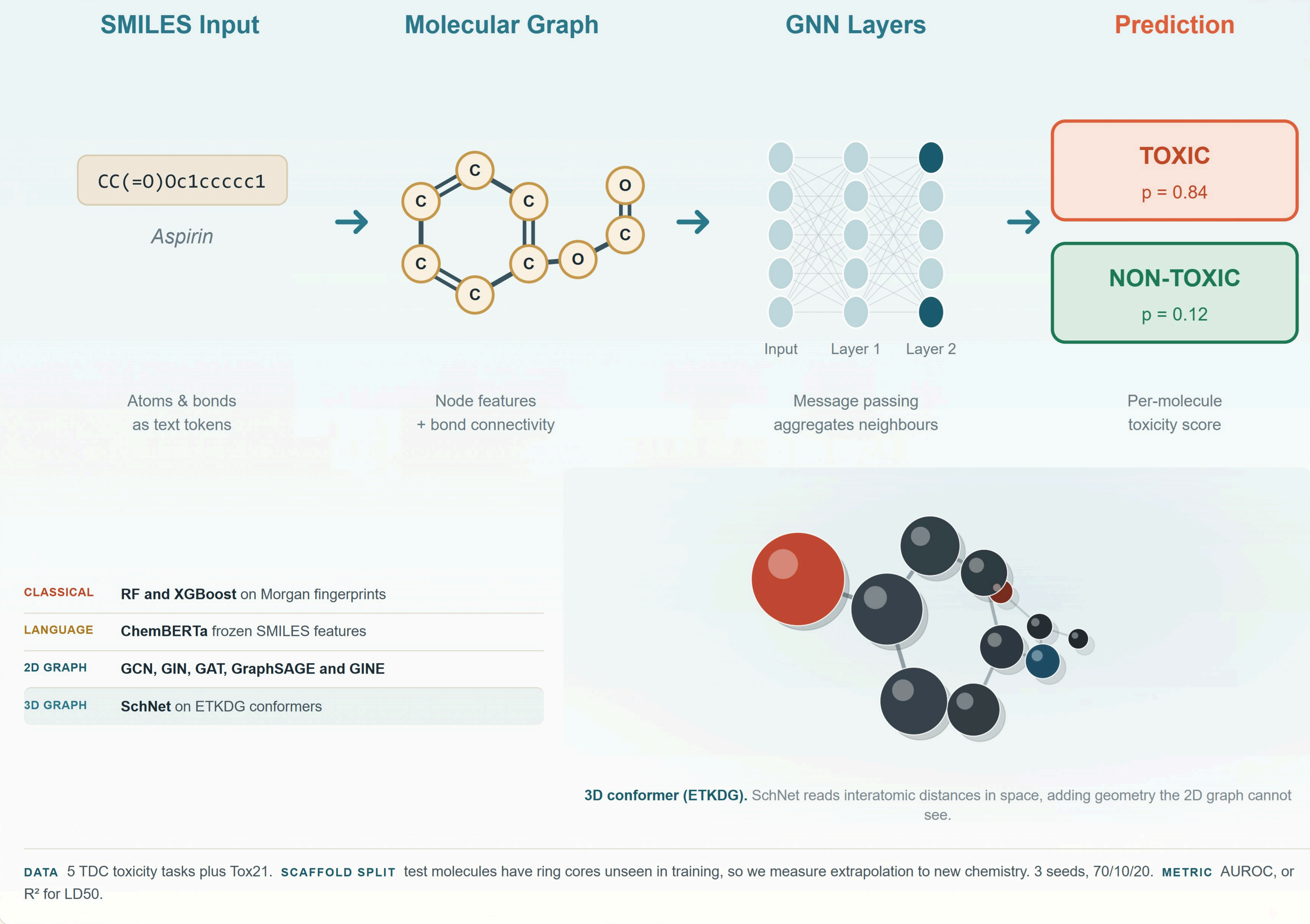
Scaffold splits reveal when graph learning actually generalizes.

GNNs help on scarce clinical tasks. Fingerprints stay strong on larger datasets. LD50 gains disappear under scaffold splits.

1 Benchmark Design

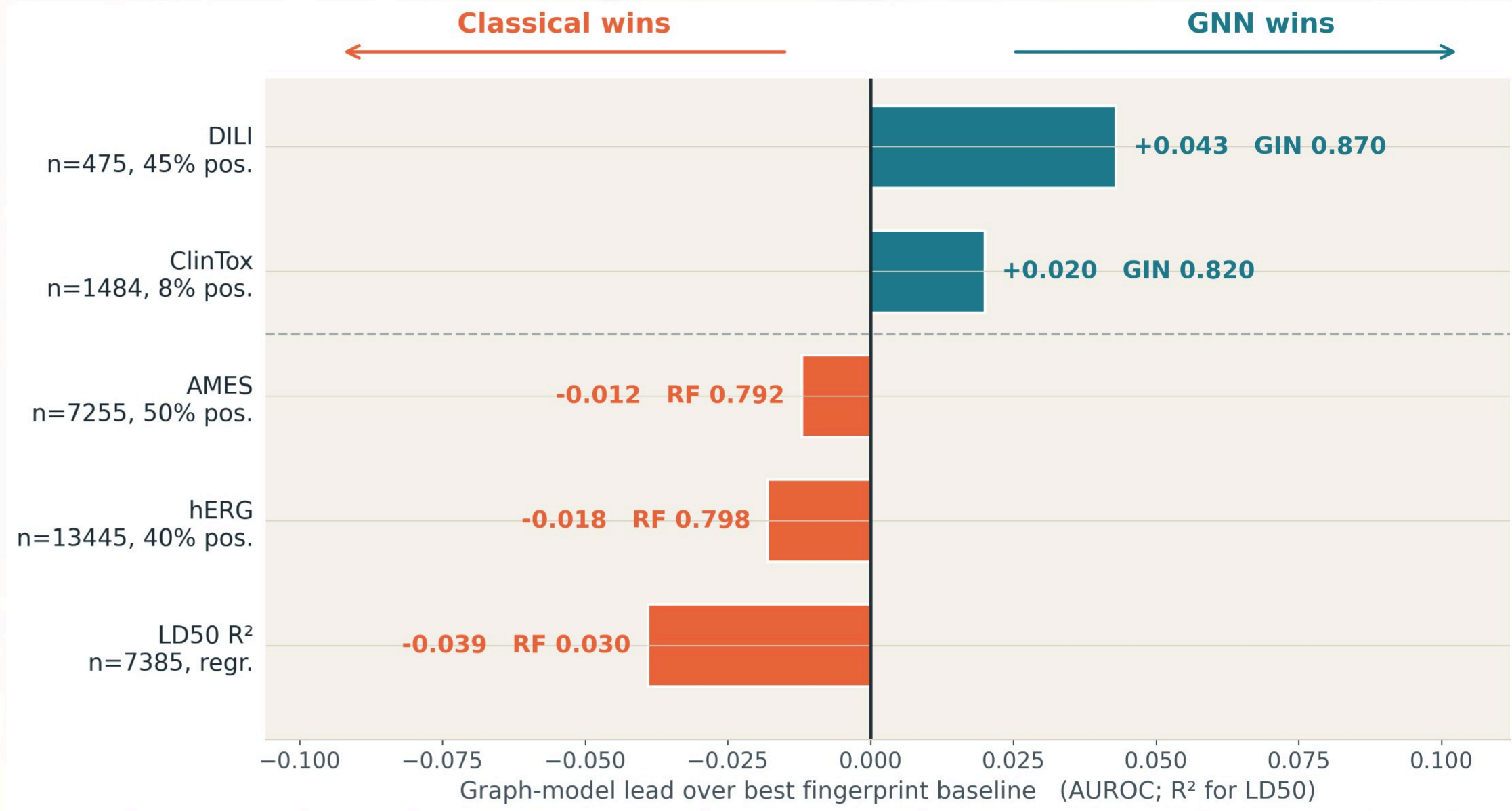
Four molecular representations, one protocol

Do graph and geometric models beat fingerprint baselines for toxicity once the evaluation is realistic? Four representation families are tested on the same molecules, scaffold split and metrics.



2 Task Regime Map

How far the best graph model leads the best fingerprint baseline



Two regimes appear. Clinical tasks favour GIN. Larger datasets favour fingerprints.

Model choice follows the task. Size, balance and chemistry shape the winner.

3 Full Benchmark

8 methods, 5 datasets, one scaffold split protocol

		Classification (AUROC)				Regression (R²)
		AMES	ClinTox	DILI	hERG	LD50 R²
Classical	RF	0.792	0.733	0.827	0.798	0.030
	XGBoost	0.757	0.800	0.815	0.794	-0.049
GNN	ChemBERTa	0.708	0.625	0.701	0.688	NA
	GCN	0.718	0.691	0.845	0.692	-0.073
	GIN	0.780	0.820	0.870	0.778	-0.523
	GAT	0.749	0.745	0.836	0.703	-0.020
	GraphSAGE	0.774	0.815	0.840	0.716	-0.009
	GINE	0.758	0.810	0.860	0.780	-0.755

Weaker Stronger per task. Bold marks the best method.

BEYOND THE HEADLINE GAP

ChemBERTa is weakest on every classification task
Frozen language-model features do not transfer to unseen scaffolds.

GIN and GINE lead the graph models
Bond features in GINE help most on the structure-sensitive endpoints.

Random Forest still wins the large datasets
Graph models lead only on the scarce, imbalanced clinical tasks.

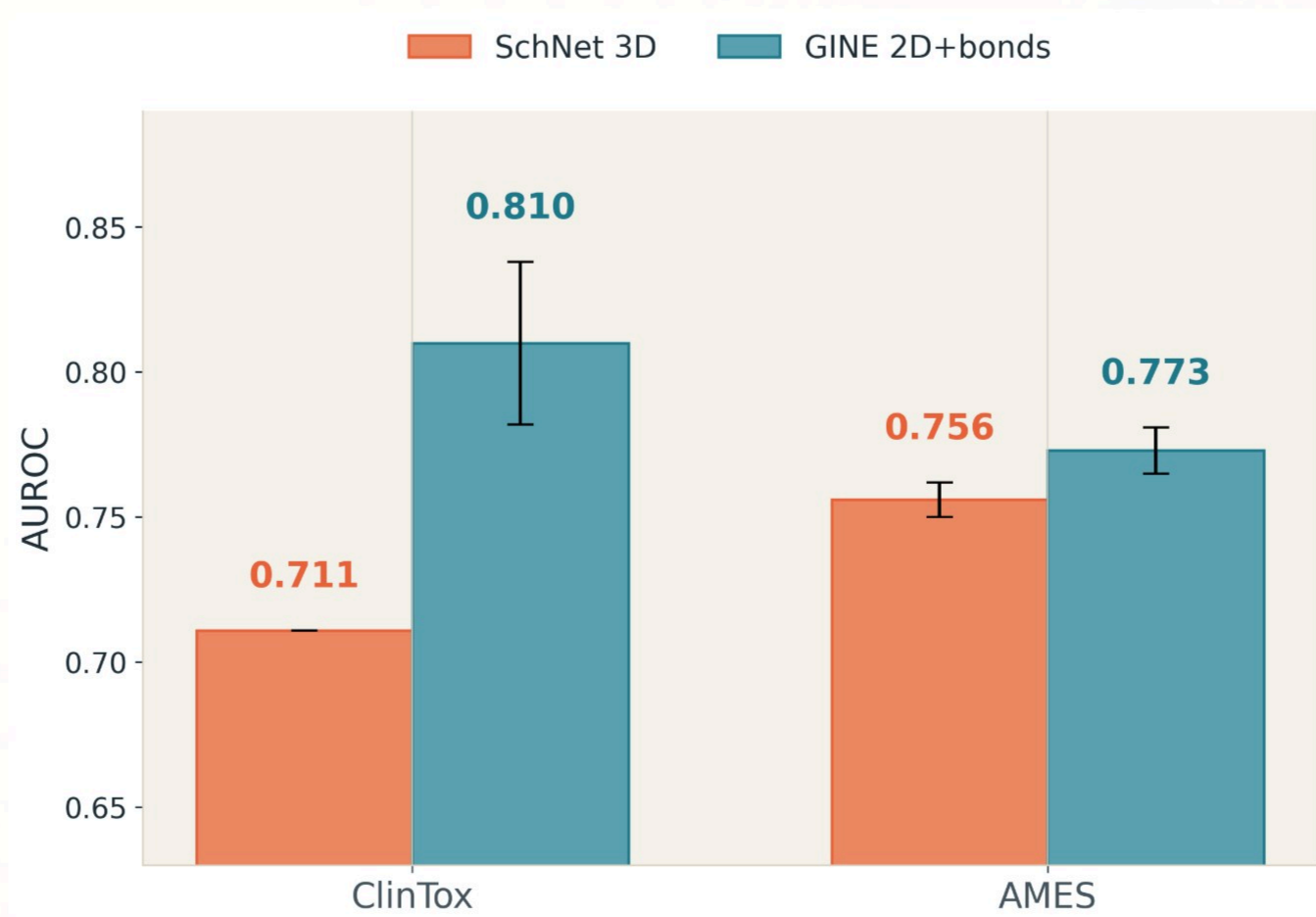
4A LD50 Split Leakage



Random splits reward near-neighbour memorization. Scaffold splits test extrapolation to unseen chemical cores.

Every GNN's LD50 R² falls below zero under scaffold splits.

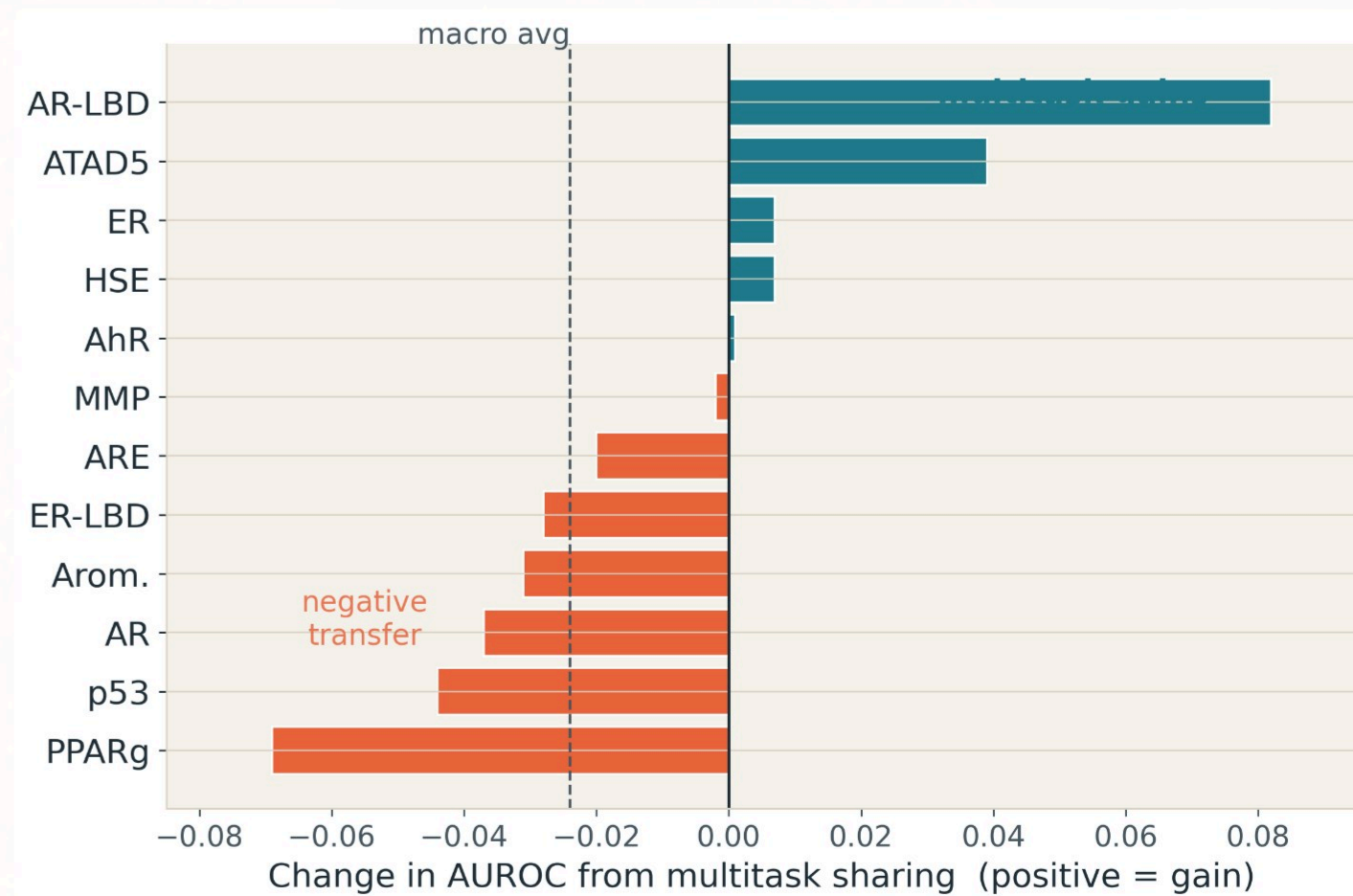
4B 3D Geometry Check



3D geometry is not automatically better. Approximate conformers add spatial signal, but also noise.

SchNet does not beat the 2D bond-aware GINE here.

4C Selective Tox21 Transfer



Sharing one GIN backbone across all 12 endpoints lowers macro AUROC. Only related endpoints benefit.

Multitask learning helps only for related endpoints.



Scan for the
interactive microsite

thekonang.github.io/AML_Microsite

5 Conclusions

- Use the task regime first. Scarce clinical endpoints suit GIN.
- Trust scaffold splits. LD50 random-split gains do not hold.
- Use 3D carefully. SchNet trails GINE but adds an uncertainty signal.
- Share tasks selectively. Tox21 transfer is not uniformly helpful.

REFERENCES

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SETUP & REPRODUCIBILITY

RDKit featurisation, PyTorch Geometric, scikit-learn and XGBoost. All runs use a shared Bemis-Murcko scaffold splitter for cross-method comparability. Per-run weights and metrics are checkpointed for reproducibility.

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